

Mastering Quantum Mechanics — Problem 9.3

Harmonic oscillator expectation value

Solved by eattardo@gmail.com

Problem 9.3

Harmonic oscillator expectation value. Calculate the expectation value of the operator $\hat{x}^4$ in the energy eigenstate with number n , and show that

$$\langle \varphi_n, \hat{x}^4 \varphi_n \rangle = \frac{1}{4} L_0^4 (6n^2 + 6n + 3).$$

Solution

Strategy: expand $\hat{x}$ in ladder operators

Following the book's recommended strategy for $\hat{x}/\hat{p}$ -dependent expectation values on energy eigenstates, write $\hat{x}$ in terms of the annihilation and creation operators (Eq. (9.3.12)):

$$\hat{x} = \frac{L_0}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger), \quad L_0 = \sqrt{\frac{\hbar}{m\omega}}, \quad [\hat{a}, \hat{a}^\dagger] = 1 \quad (\text{Eq. (9.3.10)}). \quad (1)$$

Therefore

$$\hat{x}^4 = \frac{L_0^4}{4} (\hat{a} + \hat{a}^\dagger)^4 \implies \langle \hat{x}^4 \rangle_{\varphi_n} = \frac{L_0^4}{4} \langle \varphi_n, (\hat{a} + \hat{a}^\dagger)^4 \varphi_n \rangle. \quad (2)$$

The whole problem reduces to showing $\langle \varphi_n, (\hat{a} + \hat{a}^\dagger)^4 \varphi_n \rangle = 6n^2 + 6n + 3$.

A clean trick: act with the square, then take the norm

Let $\hat{A} \equiv \hat{a} + \hat{a}^\dagger$, which is Hermitian. Then

$$\langle \varphi_n, \hat{A}^4 \varphi_n \rangle = \langle \hat{A}^2 \varphi_n, \hat{A}^2 \varphi_n \rangle = \|\hat{A}^2 \varphi_n\|^2. \quad (3)$$

Compute $\hat{A}^2$ using $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$:

$$\hat{A}^2 = (\hat{a} + \hat{a}^\dagger)^2 = \hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} = \hat{a}^2 + \hat{a}^{\dagger 2} + (2\hat{N} + 1), \quad \hat{N} \equiv \hat{a}^\dagger\hat{a}. \quad (4)$$

Acting on φ_n with the ladder relations $\hat{a}\varphi_n = \sqrt{n}\varphi_{n-1}$, $\hat{a}^\dagger\varphi_n = \sqrt{n+1}\varphi_{n+1}$ (Eq. (9.4.26)):

$$\hat{a}^2\varphi_n = \sqrt{n(n-1)}\varphi_{n-2}, \quad (5)$$

$$\hat{a}^{\dagger 2}\varphi_n = \sqrt{(n+1)(n+2)}\varphi_{n+2}, \quad (6)$$

$$(2\hat{N} + 1)\varphi_n = (2n + 1)\varphi_n. \quad (7)$$

Hence

$$\hat{A}^2\varphi_n = \sqrt{n(n-1)}\varphi_{n-2} + (2n+1)\varphi_n + \sqrt{(n+1)(n+2)}\varphi_{n+2}. \quad (8)$$

Take the norm (states are orthonormal)

Because $\varphi_{n-2}, \varphi_n, \varphi_{n+2}$ are mutually orthonormal ($\langle \varphi_k, \varphi_l \rangle = \delta_{kl}$), the cross terms drop and

$$\|\hat{A}^2 \varphi_n\|^2 = n(n-1) + (2n+1)^2 + (n+1)(n+2). \quad (9)$$

Expanding,

$$n(n-1) + (2n+1)^2 + (n+1)(n+2) = (n^2 - n) + (4n^2 + 4n + 1) + (n^2 + 3n + 2) = 6n^2 + 6n + 3. \quad (10)$$

Result

$$\boxed{\langle \varphi_n, \hat{x}^4 \varphi_n \rangle = \frac{L_0^4}{4} (6n^2 + 6n + 3)} \quad (11)$$

as required.

Checks

- **Dimensions.** L_0 has units of length, so L_0^4 matches $\hat{x}^4$; the bracket is a pure number. Good.
- **Ground state.** For $n = 0$, $\langle \hat{x}^4 \rangle = \frac{3}{4} L_0^4$. Consistent with the Gaussian ground state, for which $\langle \hat{x}^2 \rangle = \frac{1}{2} L_0^2$ (Eq. (9.4.30)) and a Gaussian satisfies $\langle x^4 \rangle = 3 \langle x^2 \rangle^2 = 3 (\frac{1}{2} L_0^2)^2 = \frac{3}{4} L_0^4$.
- **Cross-check via $\langle \hat{x}^2 \rangle$.** The same method gives $\langle \hat{x}^2 \rangle_{\varphi_n} = \frac{L_0^2}{2} (2n+1)$ (Eq. (9.4.30)); our $\langle \hat{x}^4 \rangle$ has the right large- n growth $\sim \frac{3}{2} L_0^4 n^2$, i.e. $\langle \hat{x}^4 \rangle \rightarrow \frac{3}{2} \langle \hat{x}^2 \rangle^2$ as $n \rightarrow \infty$, the expected ratio for a classical sinusoid ($\overline{x^4} / \overline{x^2}^2 = 3/2$).